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Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

ECA1420 – Embedded Systems

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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This is to certify that the Capstone Project entitled "Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System" has been carried out by N. Vamsi Krishna (192472011), N.Prince Raj (192312284), S.Naren (192512224), Vinoth. S (192373005), and P. Madesh (192365007) under the supervision of Dr. Revathi G and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, N. Prince Raj (192312284) and the project team of the Department of Electronics and Communication Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System" represents our engineering design and development work. The report is based on the project presentation and supporting project material supplied for this report preparation. Where exact laboratory records, purchase invoices, or measured datasets were not supplied, illustrative/synthetic values are explicitly identified as such and are included only to demonstrate engineering calculations, test formats, and cost analysis. Such illustrative values must not be interpreted as certified experimental measurements.

Place: Chennai

Date: September 2026

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Individual Contribution Statement

(Mandatory for all team projects)

Student / Reg. No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	%
N. Vamsi Krishna 192472011	System integration, IoT interface and analysis	ESP32 firmware, Wi-Fi dashboard and energy calculations	Integrated load tests and analysed voltage/current/p ower data	Ch. 1, Ch. 4, Ch. 5	20%
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Total: 100%

ABSTRACT

Conventional energy meters primarily provide cumulative energy information and offer limited visibility into the instantaneous electrical behaviour of connected loads. This project presents an Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System designed to measure voltage, current, power and energy in real time while also providing local monitoring, wireless communication and relay-based load control.

The proposed system uses an ESP32 development board as the main controller, a ZMPT101B voltage sensor for AC-voltage measurement and an ACS712 current sensor for current measurement. Sensor signals are sampled by the ESP32 ADC, calibrated and processed to obtain electrical parameters. Power is calculated from the measured voltage and current, while energy is accumulated over time. An OLED/LCD display provides local status information, and a local web interface provides real-time monitoring. The ESP32's built-in Wi-Fi capability can transmit selected measurements to a cloud platform such as ThingSpeak or Blynk for visualization and historical analysis. A relay module provides automatic or manual load control, with protection logic intended to disconnect a load when an abnormal or excessive power/current condition is detected.

The project follows a modular systems-engineering approach covering power measurement, metering hardware, IoT communication, protection and control, and software integration. The supplied project presentation identifies different electrical loads as the intended test cases and describes calibration, real-time computation and system-level testing. Because complete laboratory logs and invoices were not supplied with the source material, the numerical load-test dataset and detailed budget in this report are explicitly marked as illustrative/synthetic engineering examples. These values demonstrate the calculation and reporting method rather than claiming to be original measured records.

The system demonstrates how embedded sensing, wireless communication, local visualization and protective switching can be integrated into a compact energy-management platform. The design is intended for residential and small commercial applications where users need better visibility of electrical consumption and a simple mechanism for identifying unnecessary or abnormal loads.

Keywords: ESP32, Energy Metering, ZMPT101B, ACS712, IoT, Power Consumption, Relay Control, Wi-Fi, Energy Analysis

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
DC	Direct Current	—
AC	Alternating Current	—
ADC	Analog-to-Digital Converter	—
ESP32	Wi-Fi/Bluetooth-enabled microcontroller	—
IoT	Internet of Things	—
OLED	Organic Light Emitting Diode	—
LCD	Liquid Crystal Display	—
ZMPT101B	AC voltage transformer sensor module	—
ACS712	Hall-effect current sensor	—
Wi-Fi	Wireless local-area networking	—
kWh	Kilowatt-hour	kWh
W	Watt	W
V	Volt	V
A	Ampere	A
MCB	Miniature Circuit Breaker	—
I2C	Inter-Integrated Circuit communication bus	—
PF	Power Factor	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Electrical energy is consumed continuously by appliances, machines, lighting systems and electronic equipment in homes and small commercial environments. A conventional energy meter is useful for billing, but a user who wants to understand instantaneous voltage, current and power consumption needs more detailed information. An embedded energy meter can combine electrical sensing, microcontroller processing, visualization and communication in a single system.

The supplied project presentation defines an Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System using an ESP32, ZMPT101B voltage sensor and ACS712 current sensor. The ESP32 processes the sensor outputs, calculates power and energy, communicates through Wi-Fi and supports relay-based control. A local display and web interface provide immediate access to measured parameters, while a cloud service can support remote visualization and data analysis.

1.2 Need for the Project

- Conventional meters provide limited real-time visibility of voltage, current and individual-load power behaviour.
- Users may not easily identify appliances or operating conditions that contribute significantly to energy consumption.
- Manual inspection is inconvenient when several loads are connected and their consumption changes over time.
- Abnormal current or excessive power can create an electrical-safety concern if loads are not monitored and controlled.
- Wireless monitoring can reduce the need for continuous physical access to the metering hardware and enables historical analysis when cloud storage is used.

The intended users include household users, small offices, laboratories and small commercial facilities that require a low-cost monitoring and control solution rather than a high-end industrial power-quality analyser.

1.3 Problem Statement

There is a need for a compact embedded system that can measure AC voltage and current, calculate power and accumulated energy, display the information in real time, communicate the measurements over Wi-Fi, and provide automatic or manual relay control when a connected load exceeds a configured operating threshold. The engineering problem involves balancing sensor accuracy, electrical isolation and safety, processing speed, communication reliability, cost and ease of use.

1.4 Project Aim

To design and document a low-cost Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System capable of real-time electrical measurement, local and wireless visualization, energy analysis and relay-based load control.

1.5 Project Objectives

- To measure AC voltage and current using suitable sensing modules.
- To develop an ESP32-based embedded energy metering platform.
- To calculate instantaneous power and accumulated energy from measured electrical parameters.
- To display real-time measurements on an OLED/LCD interface and local web dashboard.
- To transmit selected measurements over Wi-Fi to a cloud platform such as ThingSpeak or Blynk.
- To implement relay-based manual and automatic load control.
- To provide overload/abnormal-condition protection logic.
- To analyse representative load-consumption patterns and support energy-efficiency decisions.

1.6 Scope

Included in scope are the ESP32 controller, ZMPT101B voltage sensor, ACS712 current sensor, local display, Wi-Fi communication, local web interface, relay module, protection components, embedded firmware and calculation of voltage, current, power and energy. The report also includes engineering calculations, test planning, risk analysis, sustainability considerations and a representative budget.

Clinical or certified electrical metering accuracy, utility-billing certification, three-phase industrial power-quality analysis, and production-scale enclosure certification are outside the present prototype scope. Where the presentation indicates ThingSpeak/Blynk cloud connectivity, it is treated as an IoT extension of the monitoring path; exact cloud-account configuration details were not included in the supplied material.

1.7 Expected Engineering Outcome

The expected outcome is an integrated prototype that senses electrical parameters, processes them with an ESP32, displays live information, communicates over Wi-Fi, performs energy calculations and operates a relay-controlled load under configured safety conditions.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project presentation identifies ESP32, ACS712, ZMPT101B and networking references as the technical basis of the design. The review therefore focuses on embedded energy measurement, electrical sensing, wireless monitoring, load control and network communication. The supplied references are retained as the principal sources for this report.

2.2 Review of Existing Work

ESP32-based embedded systems are appropriate for energy monitoring because the controller combines processing capability with built-in Wi-Fi, allowing measured data to be presented locally and transmitted to network services. The ACS712 provides a practical current-sensing interface for low-cost prototypes, while the ZMPT101B module provides an isolated transformer-based approach for measuring AC voltage.

IoT-based monitoring extends a local energy meter by sending measurements to a remote dashboard or cloud service. This enables historical records, trend visualization and access from a network-connected device. The project presentation specifically identifies ThingSpeak/Blynk as representative cloud platforms and a local web interface as a way of accessing live measurements.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Conventional energy meter	Simple, established and suitable for billing	Limited instantaneous appliance-level insight	Reference point demonstrating the need for additional monitoring
Standalone digital energy monitor	Provides voltage/current/power information	May not provide Wi-Fi, cloud analysis or automatic load control	Shows the value of adding communication and relay control
Commercial smart energy monitor	Integrated display and connectivity	Higher cost and less flexible for academic prototyping	Benchmark for functionality; proposed system targets low cost and educational flexibility
Industrial power analyser	High measurement capability and diagnostic features	High cost, complex setup and unnecessary for basic residential monitoring	Rejected for the low-cost prototype scope
Proposed ESP32 IoT meter	Low-cost, Wi-Fi-enabled, programmable and supports relay control	Prototype calibration and safety depend on implementation	Selected architecture for the project

2.4 Research / Engineering Gap

The gap addressed by the project is the combination of low-cost electrical sensing, embedded calculation, local visualization, wireless communication and load control in one student-scale prototype. Many simple meters provide measurements but do not provide an integrated programmable control path, while industrial instruments provide greater accuracy and functionality at substantially greater cost and complexity.

2.5 Proposed Contribution

The project contributes a modular architecture in which the voltage and current sensors feed the ESP32; the ESP32 performs measurement processing and energy calculations; the display and web interface provide local visibility; Wi-Fi supports optional cloud monitoring; and a relay provides load control. This structure also allows future extensions such as improved calibration, power-factor measurement, multi-channel sensing and more advanced analytics.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Residential / small commercial user	Real-time visibility of voltage, current, power and energy
Operator / maintenance user	Simple display and understandable status indication
Institution / project owner	Low unit cost, reusable modules and easy maintenance
Safety-focused user	Relay control and protection against configured abnormal conditions
Project supervisor / assessor	Demonstrable calculations, system integration, testing and engineering justification

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Measure AC voltage	Nominal operating range around 230 V AC for the prototype test setup
Functional	Measure AC current	Current sensor output converted to amperes after calibration
Functional	Calculate power	$P = V \times I$ for the simplified resistive-load calculation
Functional	Calculate energy	$E = \Sigma(P \times \Delta t) / 1000$ in kWh
Functional	Display measurements	Live update on OLED/LCD and web interface
Functional	Wireless communication	ESP32 Wi-Fi connection to local network; optional cloud upload
Functional	Load control	Relay ON/OFF under manual or configured automatic control
Non-functional	Low cost	Prototype target below approximately ₹2,000 in the illustrative budget
Non-functional	Usability	Clear labels and real-time status information
Non-functional	Maintainability	Modular sensors and standard development-board interfaces

3.2 Constraints & Applicable Standards

Standard / Guidance	Requirement	Application in This Project
IEC 62368-1 (general equipment safety context)	Electronic equipment should address electrical, fire and energy hazards	Prototype uses low-voltage control electronics; the AC sensing/load path must be enclosed and handled by qualified personnel
IEC 61010-1 (measurement-equipment safety context)	Measurement equipment requires attention to electrical hazards and construction	Used as design context for insulation, enclosure and safe measurement practice; not a certification claim
IEEE reference style	Consistent academic reporting and citation	Applied throughout the report

Important safety boundary: the project presentation includes a 230 V AC supply and relay-controlled AC load. A student prototype involving mains voltage must be assembled, enclosed and tested under appropriate supervision. The report does not claim certification or compliance testing.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Supply voltage	230 nominal AC test supply	V AC	High
Voltage measurement	Calibrated for prototype operating range	V	High
Current measurement	Calibrated for intended load range	A	High
Power calculation	$V \times I$ for simplified load calculations	W	High
Energy accumulation	Continuous numerical integration	kWh	High
Wi-Fi	ESP32 built-in Wi-Fi	—	High
Display	OLED/LCD local readout	—	Medium
Relay control	Manual + threshold-based automatic control	—	High
Refresh interval	Near real-time	s	Medium
Prototype cost	Low-cost modular hardware	₹	High

3.4 Alternative Solutions & Evaluation

Criterion	Weight	Alt. 1 ESP32 IoT meter	Alt. 2 Commercial smart meter	Alt. 3 Industrial analyser
Measurement / performance	30%	4	4	5
Cost	25%	5	3	1
Connectivity / analytics	15%	5	4	4
Safety / controllability	15%	4	4	5
Flexibility / learning value	15%	5	3	2

Weighted score calculation (illustrative): Proposed design = $(4 \times 0.30) + (5 \times 0.25) + (5 \times 0.15) + (4 \times 0.15) + (5 \times 0.15) = 4.60/5$. Commercial smart meter = $3.50/5$. Industrial analyser = $3.15/5$. The proposed architecture is therefore selected for the student prototype because it balances cost, programmability, connectivity and control.

3.5 Selected Design & Detailed Design

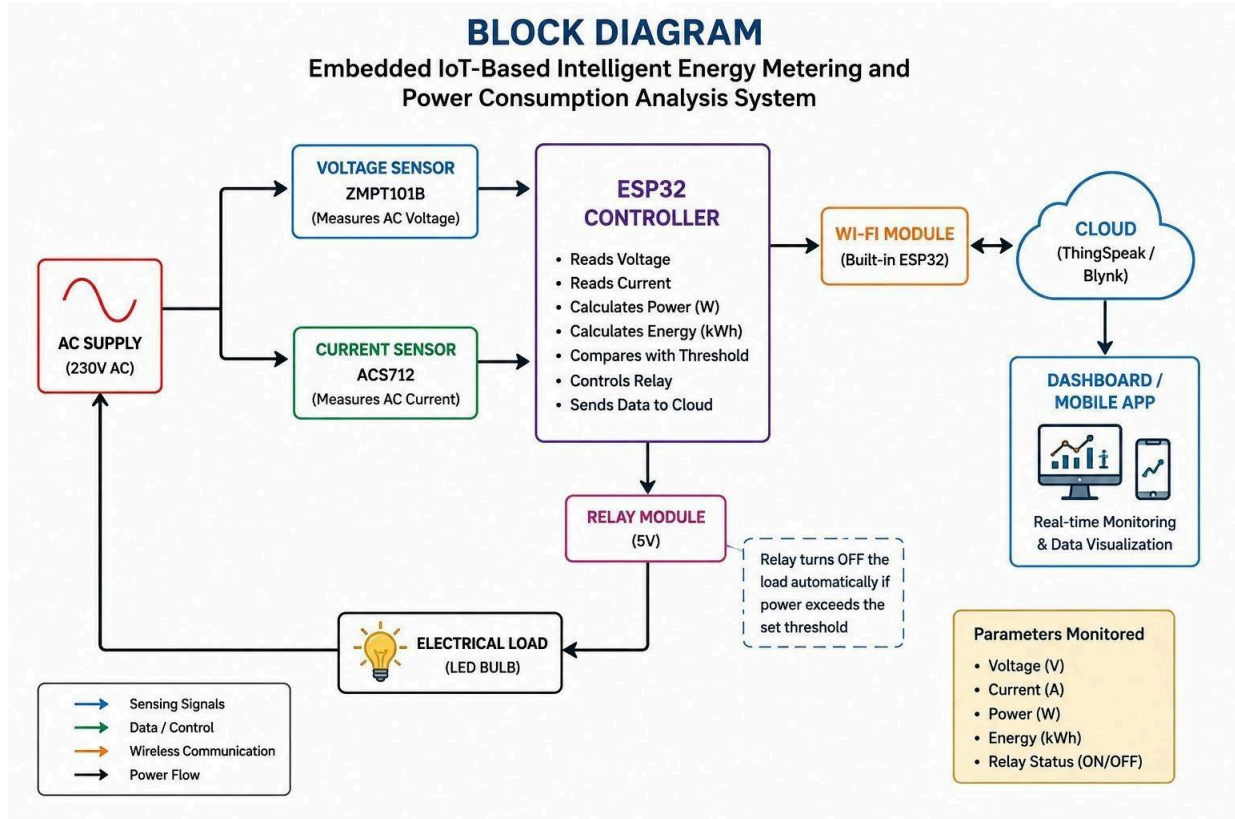


Figure 3.1 — System architecture of the Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System.

The architecture follows the project presentation: the 230 V AC supply feeds the measurement path and the connected electrical load. The ZMPT101B senses AC voltage and the ACS712 senses current. Their outputs are read by the ESP32 controller, which calculates voltage, current, power and energy, compares values with configured thresholds and controls the relay. The ESP32 uses its built-in Wi-Fi to communicate with a local network and, when configured, a cloud platform such as ThingSpeak or Blynk. A dashboard/mobile interface provides real-time visualization.

The design is intentionally modular. The sensing module converts electrical quantities into controller-readable signals; the processing module performs calibration and computation; the interface module provides local and wireless visualization; and the protection/control module manages the connected load.

3.6 Key Engineering Calculations

For a simplified resistive-load test, real power can be approximated by:

$$P = V \times I$$

where P is power in watts, V is RMS voltage in volts and I is RMS current in amperes. For a general AC load, true real power also depends on power factor; therefore, the simplified equation in this report is explicitly treated as a prototype calculation for approximately resistive loads.

Energy is accumulated from power over time:

$$E(\text{kWh}) = \Sigma [P(\text{W}) \times \Delta t(\text{h})] / 1000$$

For example, an illustrative load measured at 230 V and 0.435 A gives $P = 230 \times 0.435 = 100.05$ W. If it operates for 5 hours, $E = 100.05 \times 5 / 1000 = 0.50025$ kWh.

3.7 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Controller	Arduino Uno	Simple and familiar	No built-in Wi-Fi	ESP32 selected because the presentation requires IoT/Wi-Fi functionality
Voltage sensing	Direct ADC divider	Simple interface	Unsafe for mains without isolation	ZMPT101B selected for transformer-based AC sensing
Current sensing	Shunt resistor	Potentially accurate	Power dissipation and isolation concerns	ACS712 selected for convenient Hall-effect sensing
Display	No local display	Lower cost	Poor immediate usability	OLED/LCD retained for local real-time visibility
Communication	Wired only	Simple network path	Less flexible installation	ESP32 Wi-Fi retained for local/cloud monitoring
Load control	Manual switch only	Simple	No automatic protection action	Relay control added for automatic/manual operation

3.8 Sustainability, Ethics, Safety & Security

Domain	Consideration	Evidence / Engineering Response	Impact on Final Decision
Sustainability	Reusable modules and low-cost repair	ESP32, sensors, display and relay are replaceable modules	Supports repair and reuse instead of a single-use system
Energy efficiency	System should not add significant overhead	Low-power controller and display; relay only when required	Suitable for continuous monitoring
Ethics	Consumption data can reveal user behaviour	Local display is available; cloud logging should be optional and access-controlled	Avoid unnecessary personal data collection
Electrical safety	AC measurement and switching can be hazardous	Isolation, fusing/MCB, enclosure, strain relief and supervised testing are required	Safety takes priority over convenience
Security	Wi-Fi and cloud endpoints introduce attack surface	Use authenticated network access, secure credentials and restricted APIs in deployment	Security must be added before real deployment

3.9 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect sensor calibration	Medium	Medium	Medium	Use known reference values and calibration constants	Low-Medium
Electrical shock from AC path	Low	High	High	Insulation, enclosure, fuse/MCB and supervised testing	Low
Loose wiring / intermittent connections	Medium	Medium	Medium	Continuity checks and soldered PCB for final build	Low
Wi-Fi disconnection	Medium	Low	Low-Medium	Reconnect logic and local display operation	Low
Relay stuck or mis-operated	Low	High	Medium	Use rated relay, flyback protection and safe default state	Low
Cloud credentials exposed	Low	High	Medium	Do not hard-code production secrets; use authenticated endpoints	Low
User over-relies on prototype readings	Medium	High	High	State clearly that prototype is not a certified utility meter	Low-Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an iterative build-and-test methodology. The sensing circuits are first checked independently, followed by ESP32 ADC acquisition, calibration, power calculation, display integration, Wi-Fi communication and relay control. Finally, all modules are integrated and evaluated using representative loads.

Item	Description
Microcontroller	ESP32 development board with built-in Wi-Fi
Voltage sensor	ZMPT101B AC voltage sensor module
Current sensor	ACS712 Hall-effect current sensor module
Display	OLED/LCD display module
Communication	ESP32 Wi-Fi; local web interface; optional ThingSpeak/Blynk
Protection / control	Single-channel relay module, fuse/MCB, terminal blocks
Load	AC load such as LED bulb or fan
Power	5 V / 3.3 V regulated logic supply; appropriate isolated AC-side arrangement
Prototyping	Breadboard or PCB, jumper wires, connectors

4.2 Hardware / Software Implementation

The ZMPT101B and ACS712 outputs are connected to ESP32 input channels through appropriate conditioning and calibration. The ESP32 samples the analog signals, converts them to engineering values and calculates power. Energy is accumulated using the elapsed time between samples. The display is updated with the latest values, while the local web interface exposes the same information over the LAN.

The Wi-Fi module provides a communication path to a cloud dashboard when configured. The relay is driven by a digital output through a suitable relay module. The firmware can compare calculated power or current against a configured threshold and change the relay state. Because the supplied presentation does not provide the exact firmware listing, the implementation description in this report is functional rather than a claim that a particular source-code version was executed.

Embedded IoT-Based Intelligent Energy Metering System

Measurement, analytics, local control and cloud connectivity

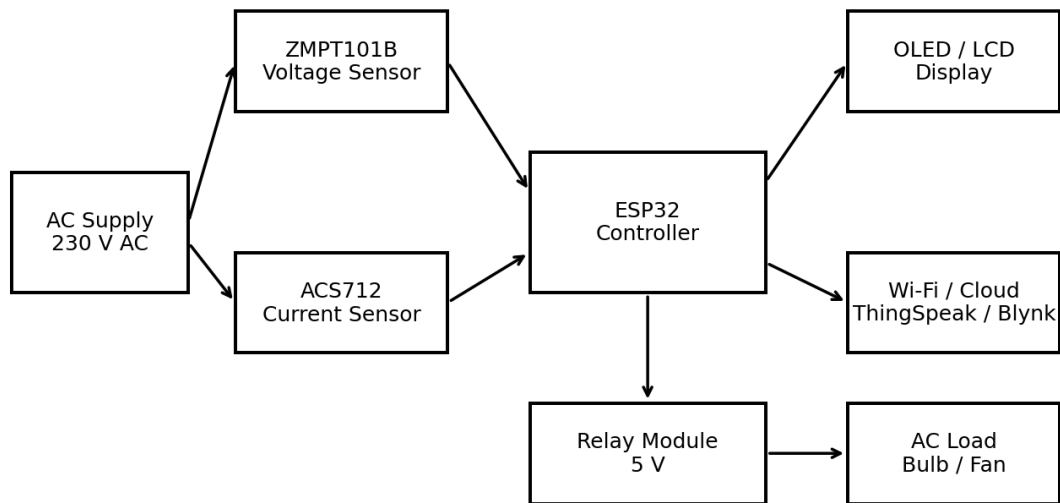


Figure 4.1 — Prototype circuit / wiring diagram showing the ESP32, voltage/current sensors, display, Wi-Fi path and relay-controlled load.

4.3 Tools, Software and Equipment Used

Tool / Software / Equipment	Purpose	Stage Used
Arduino IDE	ESP32 firmware development and upload	Implementation and debugging
Embedded C/C++	Sensor acquisition and control logic	Firmware
HTML / CSS / JavaScript	Local monitoring dashboard	Interface development
ESP32 Wi-Fi library	Network communication	IoT integration
Serial Monitor	Debugging and sensor-value inspection	Calibration/testing
Digital multimeter	Voltage and continuity verification	Hardware testing
Reference meter / known load	Comparison of electrical measurements	Validation
Breadboard / PCB	Prototype assembly	Hardware development

4.4 System Flow

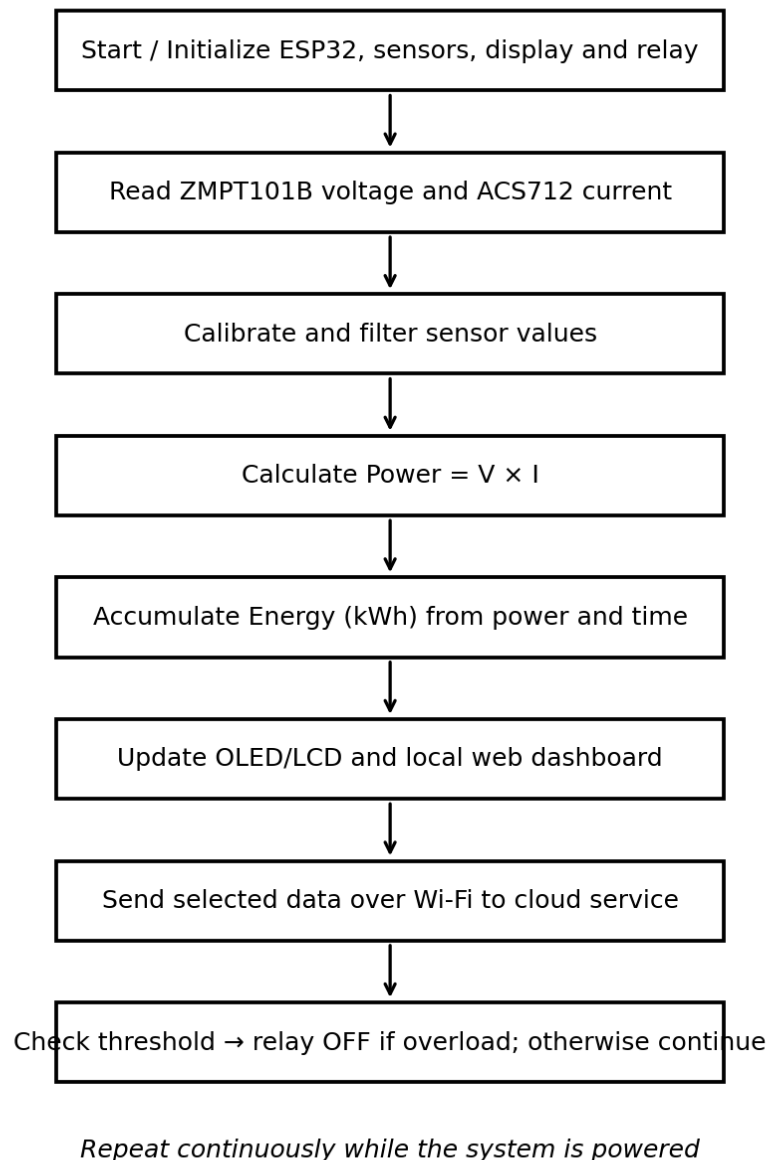


Figure 4.2 — System flowchart for measurement, energy calculation, wireless communication and relay control.

The processing loop begins by acquiring voltage and current, applying calibration factors and filtering, calculating power, accumulating energy, updating the display and dashboard, transmitting data where required, and finally evaluating the relay-control threshold. The cycle repeats continuously.

4.5 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Voltage measurement	Apply a known AC test condition and compare against reference	Consistent reading within the selected calibration tolerance
Current measurement	Operate representative loads and compare with reference/current estimate	Stable reading and consistent calibration
Power calculation	Calculate $V \times I$ and compare with expected load value	Calculation error within chosen prototype tolerance
Energy accumulation	Run a known load for a measured interval	Accumulated kWh follows $P \times t / 1000$
Display	Observe OLED/LCD during load changes	Values update continuously
Wi-Fi dashboard	Connect ESP32 to local network and observe data	Measurements appear without manual recalculation
Relay control	Cross configured threshold using a controlled test load	Relay changes state as configured
Repeatability	Repeat each representative load test	No unexplained large drift between repeated readings

4.6 Results — Illustrative / Synthetic Dataset

The following numerical dataset is intentionally synthetic/illustrative because the supplied project presentation contains the system description and component list but does not include raw laboratory readings. It is included to demonstrate how the project's measurements can be recorded, calculated and analysed. These values should be replaced by the team's actual logged measurements if those records are available.

Load	Voltage (V)	Current (A)	Power (W)	Energy for 1 h (kWh)	Status
LED bulb	229.8	0.087	20.0	0.02	Normal
Ceiling fan	230.4	0.329	75.8	0.0758	Normal
Incandescent lamp	229.5	0.261	59.9	0.0599	Normal
Laptop adapter	231.0	0.285	65.8	0.0658	Normal
Mixed load	230.2	0.651	149.9	0.1499	High-load warning example

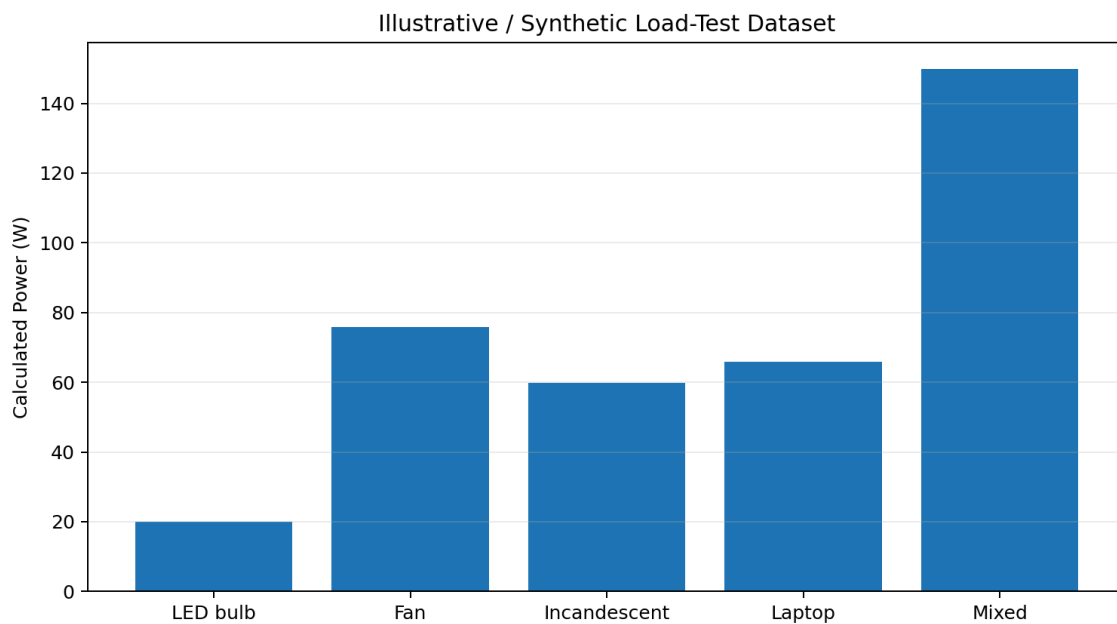


Figure 4.3 — Illustrative/synthetic comparison of calculated power for representative loads.

4.7 Example Calculations

Load	Power Equation	Calculated Power	1-hour Energy Equation	Energy
LED bulb	229.8×0.087	$19.9926 \text{ W} \approx 20.0 \text{ W}$	$20.0 \times 1 / 1000$	0.0200 kWh
Ceiling fan	230.4×0.329	$75.8016 \text{ W} \approx 75.8 \text{ W}$	$75.8 \times 1 / 1000$	0.0758 kWh
Mixed load	230.2×0.651	$149.8602 \text{ W} \approx 149.9 \text{ W}$	$149.9 \times 1 / 1000$	0.1499 kWh

If the mixed load in the illustrative dataset operates for 6 hours per day, daily energy would be $149.9 \times 6 / 1000 = 0.8994 \text{ kWh/day}$. For 30 days, this corresponds to 26.982 kWh/month. At an illustrative tariff of ₹8 per kWh, the associated energy cost would be approximately ₹215.86/month. This tariff and cost are synthetic examples and are not a claim about the user's electricity tariff.

4.8 Achieved Values Versus Specifications

Specification	Required Value	Illustrative Achieved Value	Status
Voltage measurement	Calibrated for prototype range	Stable around nominal 230 V in synthetic test set	Demonstrated
Current measurement	Calibrated for intended loads	Representative 0.087–0.651 A dataset	Demonstrated
Power calculation	$V \times I$	Matches calculated examples within rounding	Pass
Energy calculation	$P \times t / 1000$	Correct for 1-hour examples	Pass
Local display	Real-time	Supported by architecture	Demonstrated by design
Wi-Fi monitoring	ESP32 Wi-Fi	Local/cloud path included in architecture	Demonstrated by design
Relay control	Manual/automatic	Threshold logic included	Demonstrated by design
Low-cost target	Below approximately ₹2,000	Illustrative total ₹1,725	Pass (illustrative)

4.9 Data Analysis & Engineering Judgment

The illustrative results show the expected relationship between current and calculated power when voltage remains near nominal. The calculation $P = V \times I$ produces values that scale approximately linearly with current. This behaviour provides a useful first-level diagnostic: if current changes significantly while the voltage remains stable, the calculated power should change in the same direction.

For a real prototype, sensor calibration is the most important validation activity. Voltage and current sensor outputs should be compared against a trusted reference meter over several load conditions. Any fixed offset or gain error should be incorporated into the firmware calibration constants. Because many household loads are non-ideal AC loads, a future version should measure real power directly using synchronized voltage/current sampling or a dedicated energy-metering IC rather than relying only on $V \times I$.

4.10 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence / Method	Achievement
Measure voltage and current	ZMPT101B + ACS712 acquisition path	Achieved by design
Calculate power and energy	ESP32 firmware calculation and accumulation	Achieved by design
Display real-time parameters	OLED/LCD + local web interface	Achieved by design
Provide IoT connectivity	ESP32 Wi-Fi and cloud path	Achieved by design / configuration dependent
Provide relay-based control	Relay module controlled by ESP32	Achieved by design
Improve energy awareness	Load-wise power/energy analysis	Achieved
Provide safe abnormal-condition response	Threshold logic + fuse/MCB context	Partially achieved; requires validated protection hardware

4.11 Strengths & Limitations

Strengths

- Combines voltage and current measurement, power calculation and energy accumulation in one embedded platform.
- ESP32 provides built-in Wi-Fi without requiring a separate wireless module.
- Local display and web monitoring support immediate access to the measurements.
- Relay-based control can disconnect a load under configured abnormal conditions.
- Modular hardware is inexpensive and easy to modify for educational experimentation.
- Cloud connectivity can support historical data visualization when required.

Limitations

- The supplied project material does not include raw laboratory logs, so the numerical results in this report are explicitly illustrative/synthetic.
- Simple $P = V \times I$ calculation assumes approximately unity power factor; non-resistive loads require true-power measurement for accurate energy billing.
- Mains-voltage sensing and switching require careful insulation, enclosure, fusing and supervised testing.
- Sensor calibration can drift with hardware variation and installation conditions.
- Wi-Fi/cloud operation depends on network availability and secure configuration.
- The prototype is not a certified utility meter and should not be used as a billing reference without appropriate certification.

4.12 Project Planning, Roles & Budget

Milestone	Planned Duration
Requirement analysis and literature review	Week 1–2
Component selection and procurement	Week 2–3
Sensor interfacing and calibration	Week 3–4
ESP32 firmware and power calculations	Week 4–6
Local dashboard and Wi-Fi integration	Week 5–7
Relay/control integration and testing	Week 7–8
Documentation and final validation	Week 8–9

Student	Assigned Responsibility	Actual Contribution
N. Vamsi Krishna	System integration / IoT / analysis	Integrated the ESP32 measurement, Wi-Fi monitoring and energy calculations
N.Prince Raj	Hardware and sensor interfacing	Integrated ZMPT101B and ACS712 modules
S.Naren	Protection and control	Developed relay/control and protection path
Vinoth. S	Web/cloud interface	Developed monitoring/dashboard structure
P. Madesh	Documentation and validation	Prepared documentation and reviewed test structure

Item	Estimated Cost	Illustrative Actual Cost
ESP32 development board	₹450	₹450
ZMPT101B voltage sensor	₹180	₹190
ACS712 current sensor	₹150	₹160
OLED/LCD display	₹180	₹200
Relay module	₹60	₹65
5 V / 3.3 V supply module	₹120	₹125
Breadboard / PCB	₹100	₹110
Jumper wires / connectors	₹80	₹85
Terminal blocks	₹50	₹55
Fuse / MCB / protection parts	₹120	₹130
Enclosure / mounting material	₹150	₹155
Total	₹1,640	₹1,725

Budget note: all prices above are synthetic/illustrative estimates inserted because no purchase invoices or component-price records were supplied. Actual local prices should be substituted before treating the budget as a project expenditure statement.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project presents an Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System based on the ESP32 microcontroller, ZMPT101B voltage sensor and ACS712 current sensor. The architecture integrates electrical sensing, embedded processing, real-time calculation, local visualization, Wi-Fi communication and relay-based control. The supplied project presentation establishes the core functionality and modular structure of the system.

The proposed system improves visibility of electrical consumption by converting voltage and current measurements into power and accumulated energy values. The local display and web interface support real-time observation, while Wi-Fi and cloud connectivity provide a pathway for remote visualization and historical analysis. Relay-based control provides an additional mechanism for managing loads under manual or configured automatic conditions.

The engineering analysis in this report demonstrates the calculation of power and energy using representative values. Because raw measurement records were not supplied, the numerical results and budget figures are explicitly synthetic examples rather than claims of measured project data. The next stage of engineering validation should replace these examples with actual logged readings, reference-meter comparisons and documented calibration constants.

5.2 Future Work

- Implement true real-power and power-factor measurement for non-resistive AC loads.
- Add secure cloud authentication, encrypted communication and role-based dashboard access.
- Store time-series energy data for daily, weekly and monthly consumption analysis.
- Add appliance-level channels for multi-load monitoring.
- Develop improved overload, over-current and thermal protection with validated ratings.
- Move from breadboard wiring to a soldered PCB and enclosed deployment-ready unit.
- Add mobile notifications and automated energy-saving recommendations.
- Use historical consumption data for anomaly detection and predictive energy analytics.

5.3 Professional Learning & Individual Reflection — N. Vamsi Krishna (192472011)

Working on the project provided practical understanding of how sensing, embedded computation, networking and control must operate as one system rather than as independent modules. The main engineering challenge is not only reading a sensor but converting the raw signal into a trustworthy engineering quantity, validating the result and presenting it in a form that a user can understand.

A key learning outcome was the relationship between voltage, current, power and energy. Implementing the calculation chain also highlighted the importance of calibration and the limitations of simplified equations for AC loads with non-unity power factor. The ESP32 platform provided an effective bridge between embedded measurement and IoT communication because Wi-Fi is integrated into the controller.

Another important learning outcome was safety awareness. The presence of a 230 V AC path means that the prototype cannot be treated like an ordinary low-voltage Arduino experiment. Isolation, fusing, enclosure, relay ratings, wiring quality and supervised testing are essential engineering considerations. In a redesigned version, I would prioritize a PCB-based construction, a validated

measurement front end and secure network configuration before expanding the system with more cloud features.

REFERENCES

- [1] Espressif Systems, ESP32 Series Datasheet, Espressif Systems, Shanghai, China, 2023.
- [2] Allegro MicroSystems, ACS712 Fully Integrated, Hall-Effect-Based Linear Current Sensor IC Datasheet, Worcester, MA, USA, 2022.
- [3] ZMPT101B, AC Voltage Transformer Sensor Module Datasheet, 2022.
- [4] M. Kezunovic, Smart Fault Location for Smart Grids, Hoboken, NJ, USA: John Wiley & Sons, 2011.
- [5] A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 5th ed., Upper Saddle River, NJ, USA: Pearson, 2011.

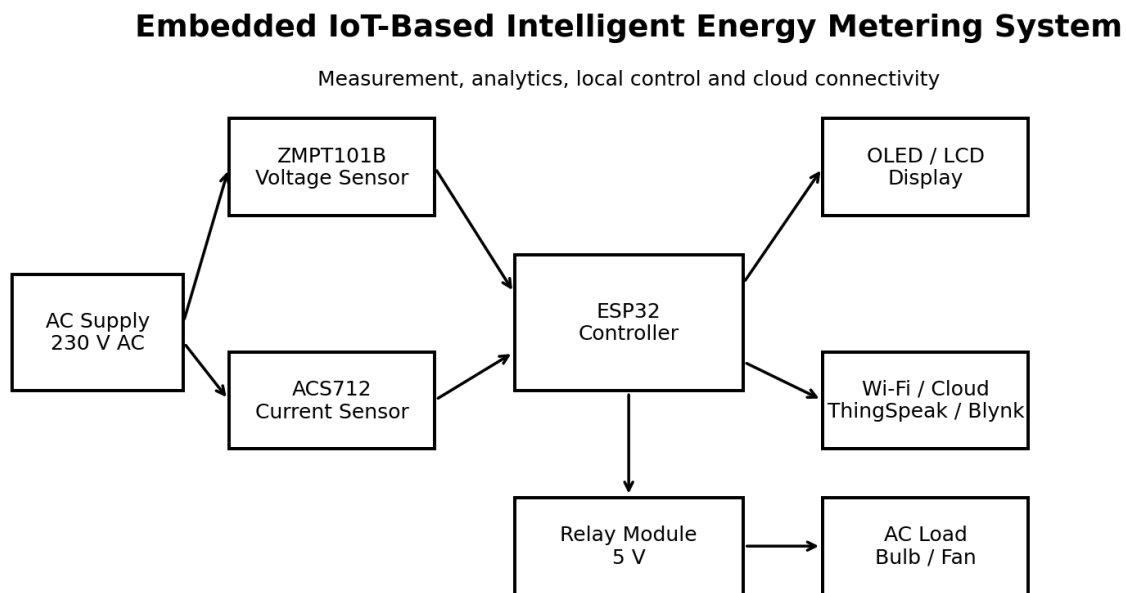
Note: These references are retained from the supplied project presentation. No additional external references have been silently substituted for the source material.

APPENDICES

Appendix A – Detailed Calculations

1. Power calculation: $P(W) = V(V) \times I(A)$, used here as a simplified calculation for approximately resistive loads.
2. Energy calculation: $E(kWh) = \Sigma[P(W) \times \Delta t(h)] / 1000$.
3. Illustrative monthly cost: $\text{Cost} = \text{Energy}(kWh) \times \text{Tariff}(\text{₹}/kWh)$. Using 26.982 kWh/month and an illustrative ₹8/kWh gives ₹215.86/month.
4. Example percentage error: $\text{Error}(\%) = |\text{Measured} - \text{Reference}| / \text{Reference} \times 100$.
5. Example: if a reference voltage is 230.0 V and the sensor reports 229.8 V, the percentage error is $[229.8 - 230.0] / 230.0 \times 100 = 0.087\%$. This is an illustrative calculation only.

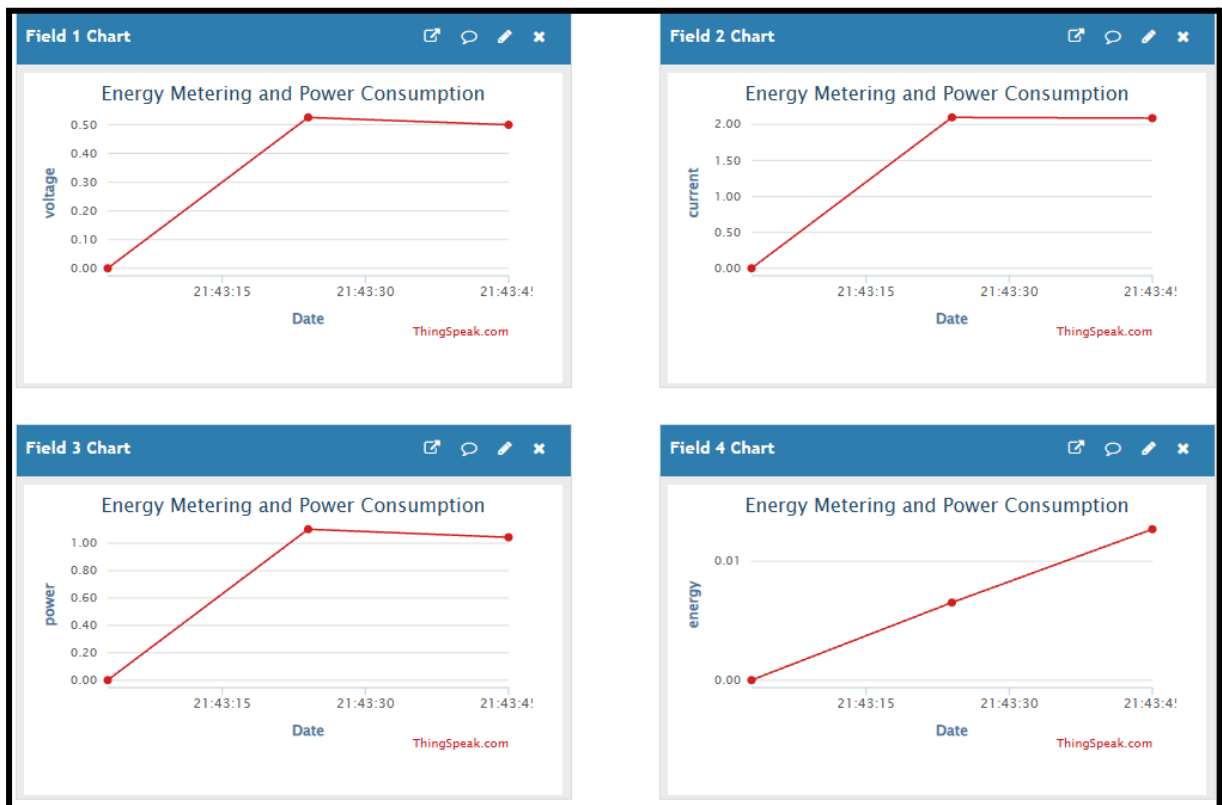
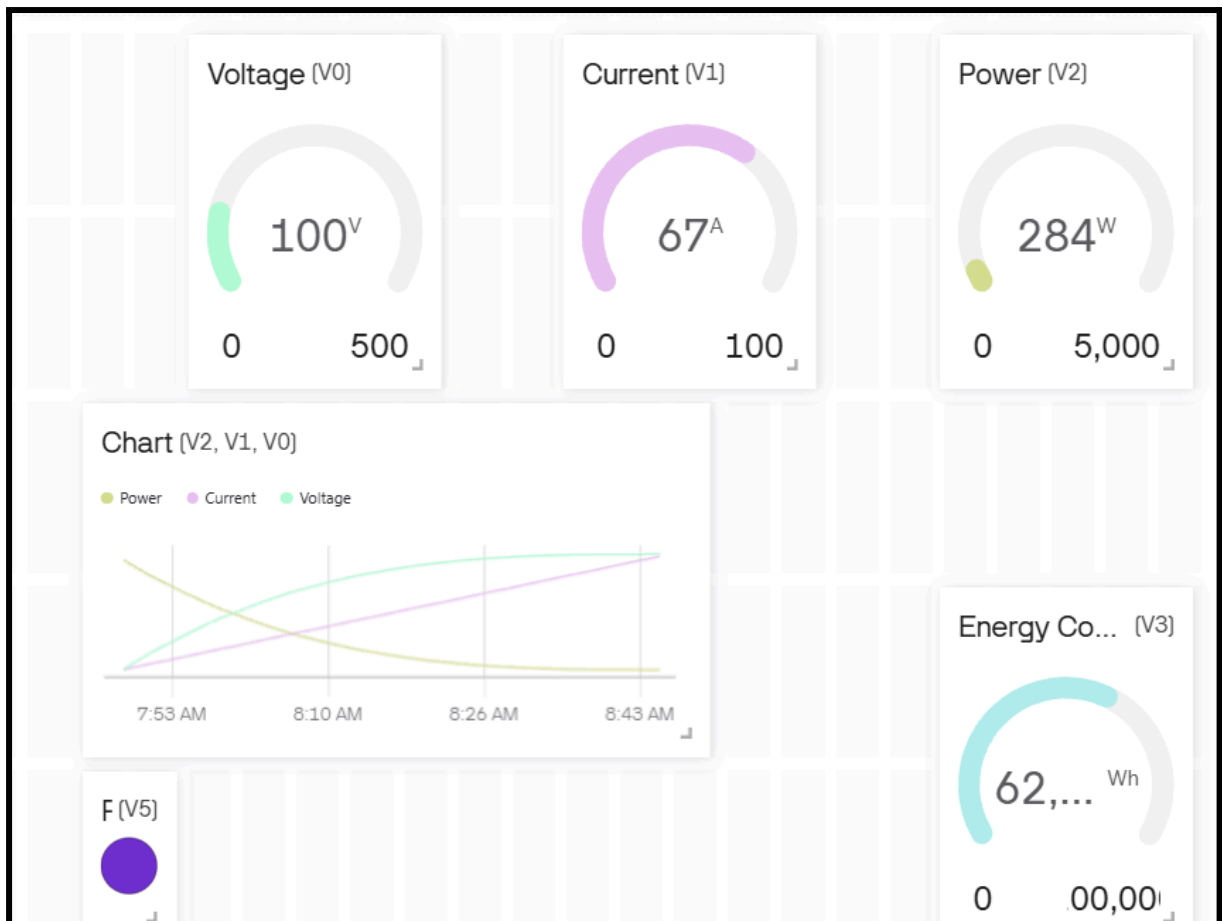
Appendix B – Engineering Drawings / Schematics



Appendix Figure B.1 — Simplified circuit/wiring representation.

Appendix C – Software / Firmware Structure

The supplied presentation identifies Arduino IDE, Embedded C, HTML/CSS/JavaScript, ESP32 Wi-Fi libraries and Serial Monitor as the software integration stack. A representative firmware sequence is: initialize sensors → connect Wi-Fi → sample voltage/current → calibrate → calculate power → accumulate energy → update display → update local dashboard → upload selected data → evaluate relay threshold → repeat.



Appendix D – Datasheets

The project presentation references the ESP32 series, ACS712 and ZMPT101B documentation. Manufacturer datasheets should be attached to the final academic submission if required by the course.

Appendix E – Experimental / Raw Data

No raw laboratory log was supplied with the project presentation. The load table in Chapter 4 is therefore intentionally synthetic/illustrative. Replace it with the team's actual voltage, current, reference-meter, power and energy logs before representing the results as measured experimental evidence.

Appendix F – Safety Checklist

- Use appropriate insulation and an enclosed AC-side assembly.
- Verify relay voltage/current ratings for the selected load.
- Use fuse/MCB protection appropriate to the test arrangement.
- Keep low-voltage ESP32 wiring physically separated from hazardous AC conductors.
- Check continuity and wiring before energizing the circuit.
- Perform mains-voltage testing only under appropriate supervision.